

Statistics

Part 8 - Point Estimation

2025 Lectures

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Recommended reading (Part 8)

For the next lecture, please, read

- sections 10.1, 10.2, 11.1, ..., 11.6 from Bartoszynski and Niewiadomska-Bugaj
- sections 7.1, ... 7.4, 8.1, 8.2, 8.3, 8.4 from Mittelhammer

Statistical inference and data I

- What can we **infer** about the underlying **stochastic mechanisms** governing the occurrence of our data?
- Any **inference** drawn from the observed data is subject to **error**. This error - crucially, not a “mistake” - is at the heart of statistical analysis.
- In the language of decision theory, we find ourselves in one of several possible scenarios, denoted as “world θ_1 ” . . . “world θ_n ,” without knowing which one accurately represents reality. Each scenario corresponds to a specific action that would be most appropriate if we knew we were in that world. To determine the actual scenario, we conduct experiments.

Statistical inference and data II

- Let X represents the results of these experiments. We assume that X follows a probability distribution dependent on the true θ_j .
- However, the outcome of X is often possible under multiple θ_j 's, albeit with varying probabilities depending on the true state of the world.
- Going forward: we assume that data are observations collected through a **simple random sampling** method.

Estimation

- **Estimation** is the process of extracting information about the value of a population parameter θ .
- An **estimator** of θ is then a **rule** (function) that provides an approximation of θ based on a sample $X_1, \dots, X_n$. It is important to note that there can be multiple estimators for the same parameter.
- We observe independent random variables $X_1, \dots, X_n$, which are sampled from a distribution that depends on θ , where θ takes values from a parameter space Θ .
- For example, $X_1, \dots, X_n$ might follow a normal distribution with an unknown mean θ and known standard deviation.
- In simple cases, θ is a scalar value, so Θ is a subset of the real line $\mathbb{R}$. However, in more complex situations, Θ can represent a multidimensional parameter space.

Unless stated otherwise, we will assume that θ is a scalar and that Θ is an interval on the real line.

Estimators

Definition 39: A statistics is called an **estimator** if it is used to estimate θ . The value of an estimator, obtained from a particular sample, is called an **estimate** of θ .

Example 82: Let $X_1, \dots, X_n \sim U[0, \theta]$. Estimate θ !

- $T_1 = X_{n:n}$... the largest value should always satisfy $X_{n:n} \leq \theta$
- $T_2 = \frac{n+1}{n}T_1$... $X_1, \dots, X_n$ divides $[0, \theta]$ into $n + 1$ intervals of “even” length
- $T_3 = (n + 1)X_{1:n}$
- $T_4 = 2\bar{X}$... average should be close to midpoint
- and the list could go on

Desired properties of estimators

Definition 40: Let $T_n = T_n(X_1, \dots, X_n)$ be the estimator. Then it is called **(weakly) consistent** if $T_n \xrightarrow{P} \theta$. It is called **strongly consistent** if $T_n \xrightarrow{a.s.} \theta$.

- “Increasing the sample size improves the accuracy of an estimator, bringing it closer to the true value of the parameter.”

Definition 41: Let $T_n = T_n(X_1, \dots, X_n)$ be the estimator. Then it is called **unbiased** if $E_\theta(T_n) = \theta$ for every n . Otherwise it is called **biased**. The difference $B_\theta(T_n) = E_\theta(T_n) - \theta$ is called **bias** of T_n . It is called **asymptotically unbiased** if $\lim_{n \rightarrow \infty} E_\theta(T_n) = \theta$.

- “If we repeat the sampling, the estimate will, on average, equal the true value of the parameter.”
- Additionally, we may require other properties, such as the estimator having the lowest variance (or “highest precision”).

Example cont.

Example 82 cont.:

- $P(T_1 \leq t) = P(X_1 \leq t, \dots, X_n \leq t) = \left(\frac{t}{\theta}\right)^n$ and thus

$$P(|T_1 - \theta| < \varepsilon) = 1 - \left(\frac{\theta - \varepsilon}{\theta}\right)^n \rightarrow 1$$

and so T_1 is consistent.

- $T_2 = \frac{n+1}{n}T_1$ and so $T_2 \xrightarrow{P} \theta$. Also T_2 is consistent.
- $P(|T_3 - \theta| < \varepsilon) = P\left(X_{1:n} < \frac{\theta+\varepsilon}{n+1}\right) - P\left(X_{1:n} < \frac{\theta-\varepsilon}{n+1}\right) =$
 $= \left(1 - \frac{\theta-\varepsilon}{\theta(n+1)}\right)^n - \left(1 - \frac{\theta+\varepsilon}{\theta(n+1)}\right)^n \rightarrow e^{-\frac{\theta-\varepsilon}{\theta}} - e^{-\frac{\theta+\varepsilon}{\theta}} < 1$
Thus T_3 is not consistent estimator of θ .

- By LLN $2\bar{X} \xrightarrow{P} 2E(X) = 2\frac{\theta}{2} = \theta$.

Mean square error

Definition 42: Let T be an estimator of θ . Then

$$MSE_{\theta}(T) = E_{\theta}[(T(X_1, \dots, X_n) - \theta)^2]$$

is called a **mean squared error** of T .

Theorem 57: For any estimator T of parameter θ we have

$$MSE_{\theta}(T) = Var_{\theta}T + (B_{\theta}(T))^2.$$

Example 82 cont.:

$$E(T_1) = \frac{n}{n+1}\theta \neq \theta, \quad E(T_2) = E(T_3) = E(T_4) = \theta,$$

$$Var(X_{1:n}) = Var(X_{n:n}) = \frac{n\theta^2}{(n+1)^2(n+2)}.$$

$$MSE_{\theta}(T_1) = \frac{2\theta^2}{(n+1)(n+2)}$$

$$MSE_{\theta}(T_2) = \frac{\theta^2}{n(n+2)}$$

$$MSE_{\theta}(T_3) = \frac{n\theta^2}{n+2}$$

$$MSE_{\theta}(T_4) = \frac{\theta^2}{3n}$$

Why $T = \bar{X}_n$ as estimate of the mean?

Assume that the observations $X_1, X_2, \dots$ are iid random variables, with $E(X_i) = \theta$ and $Var(X_i) = \sigma^2 < \infty$, where σ is assumed to be known.

- The estimator $T = \bar{X}_n$ of parameter θ is unbiased. Moreover, it is consistent and has $MSE_\theta(T) = \frac{\sigma^2}{n}$. Importantly, $MSE_\theta(T)$ for a fixed n does not depend on the parameter θ .
- It decreases only in proportion to $\frac{1}{n}$! Decrease in proportion to $\frac{1}{n^2}$ is more desirable. However, the former case is far more common in practice.
- Additionally, T is linear!

Test of consistency

Test of consistency (using Chebyshev's inequality)

- 1) Check whether the estimator T is unbiased or not.
- 2) Calculate $VarT$ and $B(T)$, the bias of T .
- 3) Conditions for consistency:
 - an unbiased estimator is consistent if $VarT \rightarrow 0$ as $n \rightarrow \infty$.
 - a biased estimator is consistent if both $VarT \rightarrow 0$ and $B(T) \rightarrow 0$ as $n \rightarrow \infty$.

Example 83: Let $X_1, \dots, X_n$ be a random sample with true mean μ and finite variance. Then, the sample mean $\bar{X}$ is a consistent estimator of the population mean μ .

Fisher information

Assume the sequence $X_1, X_2, \dots$ follows a distribution where the set of points for which $f(x, \theta) > 0$ does not depend on θ . Thus, no single observation can completely rule out any value of θ . (This excludes situations like those in Example 82). This property is referred to as the **regularity assumption**. The **amount of information about θ contained in the event $\{X = x\}$** can be quantified using the Fisher information.

Definition 43: Let X be a random variable with twice differentiable function $f(x, \theta)$ determining the distribution of X such that the set of x with $f(x, \theta) > 0$ is the same for all θ . Then the **Fisher information** about θ in a single observation X is defined by

$$I(\theta) = E_{\theta}[(J(X, \theta))^2], \text{ where } J(X, \theta) = \frac{\partial}{\partial \theta} \log f(X, \theta)$$

provided the expectation exists.

Examples

Example 84: $X \sim N(\theta, \sigma^2)$

$$J(X, \theta) = \frac{\partial}{\partial \theta} \left(-\log \sigma \sqrt{2\pi} - \frac{(X - \theta)^2}{2\sigma^2} \right) = \frac{X - \theta}{\sigma^2}$$

$$I(\theta) = \frac{1}{\sigma^4} E[(X - \theta)^2] = \frac{1}{\sigma^2}.$$

Example 85: Bernoulli trial with probability θ

$$P(X = 1|\theta) = \theta, P(X = 0|\theta) = 1 - \theta$$

$$f(x, \theta) = \theta^x (1 - \theta)^{1-x}, \quad x = 0, 1$$

$$J(X, \theta) = \begin{cases} -\frac{1}{1-\theta} & X = 0; \\ \frac{1}{\theta} & X = 1; \end{cases} \quad I(\theta) = \frac{1}{\theta(1-\theta)}.$$

So $I(\theta)$ has minimal value 4 at $\theta = \frac{1}{2}$ and for θ approaching 0 or 1 the information goes to infinity.

Properties of Fisher information

Theorem 58: Under regularity assumptions,

- a) $E_{\theta}(J(X, \theta)) = 0$;
- b) $Var_{\theta}J(X, \theta) = I(\theta)$;
- c) $I(\theta) = -E_{\theta} \left(\frac{\partial}{\partial \theta} J(X, \theta) \right)$;
- d) The information $I_n(\theta)$ in a random sample of n observations is

$$I_n(\theta) = nI(\theta).$$

Theorem 59: (Rao - Cramér)

Under regularity assumptions, for any unbiased estimator T_n of a parametric function $m(\theta)$ we have

$$Var_{\theta}T_n \geq \frac{(m'(\theta))^2}{nI(\theta)}.$$

For $m(\theta) = \theta$, i.e. T_n unbiased estimator of θ , $Var_{\theta}T_n \geq \frac{1}{nI(\theta)}$.

Efficiency of estimators

Definition 44: Any unbiased estimator T that satisfies the regularity assumption and whose variance attains the Rao-Cramér bound is called **efficient**. The ratio

$$\frac{\left(\frac{1}{nI(\theta)}\right)}{\text{Var}_{\theta}T_n} \leq 1$$

is called **efficiency**.

If $\tilde{T}_n$ and $\hat{T}_m$ are two unbiased estimators of θ ,

$$\frac{\text{Var}_{\theta}\tilde{T}_n}{\text{Var}_{\theta}\hat{T}_m}$$

is called **relative efficiency** of $\hat{T}_m$ with respect to $\tilde{T}_n$.

Method of Moments Estimators

- A method for estimating population parameters, such as the mean, variance, or median (not necessarily moments), involves equating sample moments to the corresponding unobservable population moments.
- The estimator is obtained by solving the resulting equation with respect to the unknown parameter.
- Applicability: Suitable for estimating multiple parameters simultaneously. In such cases, we solve system of corresponding equations.
- Advantages: Quick and straightforward, easily computable by hand.
- Disadvantages: Solutions may not be unique, depending on the equations chosen.
- Practical Rule: Use moments of the lowest orders to simplify the estimation process.

Method of Moments Estimators

Example 86: $X_1, \dots, X_n \sim EXP(\theta)$

The expectation of X_i is given by $E(X_i) = \frac{1}{\theta}$, and the sample mean $\bar{X}_n$ serves as a logical estimator of $E(X_i)$. Therefore, the estimator for θ based on the sample mean is

$$T_1 = \frac{1}{\bar{X}_n}.$$

Alternatively, the second ordinary moment of X_i is given by $E(X_i^2) = \frac{2}{\theta^2}$, and its sample counterpart is $\frac{1}{n} \sum_{i=1}^n X_i^2$. Thus, the estimator for θ based on the second ordinary moment is

$$T_2 = \sqrt{\frac{2n}{\sum_{i=1}^n X_i^2}}.$$

Furthermore, if we want to estimate, e.g., $p = P(X \geq 3) = e^{-3\theta}$

we can use either $p_1 = \exp\left\{-\frac{3}{\bar{X}_n}\right\}$ or $p_2 = \exp\left\{-3 \cdot \sqrt{\frac{2n}{\sum X_i^2}}\right\}$.

Both estimators provide an estimate for p , based on different moments of the sample data.

Method of Moments Estimators I

Example 87:

Consider the case where $\theta = (\mu, \sigma^2)^\top$, with both parameters unknown. Let $X_1, \dots, X_n$ be a random sample with $E(X_i) = \mu$ and $Var X_i = \sigma^2$.

Since the first moment $E(X) = \mu$ provides the basis for estimator for μ and $E(X^2) = \sigma^2 + \mu^2$, we have to solve the following two equations

$$\begin{aligned}\bar{X} &= \hat{\mu} \\ \frac{1}{n} \sum X_i^2 &= \hat{\sigma}^2 + \hat{\mu}^2\end{aligned}$$

to obtain $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$.

Method of Moments Estimators II

- Moment estimators are consistent under mild conditions. If the parameter θ is a continuous function of the moments (which is typically the case), then the moment estimators will converge to the true parameter values as the sample size n increases.
- In many cases, the moment estimators coincide with the maximum likelihood estimators. However, in certain situations, moment estimators can be inferior to MLEs.

Maximum Likelihood Estimators

- The likelihood function represents the chances of observing a particular sample $x_1, x_2, \dots, x_n$ given the parameter θ .
- Formally, if $X_1 = x_1, \dots, X_n = x_n$, are observations from a random sample, then $f(x_1, \theta) \cdots f(x_n, \theta)$, the joint probability density function or joint probability mass function of the sample, depending on the nature of the data, considered as a function of the parameter θ , is known as the **likelihood function of the sample**

$$L(\theta, x) = \prod_{i=1}^n f(x_i, \theta).$$

Definition 45: Given the sample $x = (x_1, \dots, x_n)$, the value $\hat{\theta}(x)$ maximizing $L(\theta, x)$, is called the **maximum likelihood estimate** (MLE) of θ .

- We often define $\ell(\theta, x) = \log L(\theta, x)$ and maximize $\ell(\theta, x)$ instead $L(\theta, x)$.

Maximum Likelihood Estimators

Example 88:

Suppose we observed three successes and two failures in five Bernoulli trials with probability of success θ . The probability mass function for a single trial is $f(x, \theta) = \theta^x(1 - \theta)^{1-x}$ where $x = 0$ represents failure and $x = 1$ success. The likelihood function for observing three successes and two failures is the product of the individual probabilities for each outcome

$$L(\theta, x) = \theta^3(1 - \theta)^2 \text{ for } 0 \leq \theta \leq 1.$$

The maximum likelihood estimate of θ is $\hat{\theta} = 3/5$.

Example 89:

Suppose we have two observations $x_1 = 3, x_2 = -2$ from a normal distribution $N(0, \theta^2)$. Substituting $x_1 = 3$ and $x_2 = -2$ to the log-likelihood function for this normal distribution, we get

$$\ell(\theta, x) = -\log 2\pi - 2 \log \theta - \frac{13}{2\theta^2}$$

and we derive the maximum likelihood estimate $\hat{\theta}_{MLE} = \sqrt{\frac{13}{2}}$.

Maximum Likelihood Estimators I

- The likelihood function can indeed be regarded as a random function because its values depend on the sample drawn from the population. Specifically, the likelihood function reflects the probability of observing a particular sample given a parameter θ , and since the sample is random, so is the likelihood function.
- **Invariance principle:** If $\hat{\theta}$ is the MLE of the parameter θ , then for any function $h(\cdot)$, the estimator $h(\hat{\theta})$ is the MLE of parameter $h(\theta)$. This principle is particularly useful in transformation problems.
- **Example:** If $\hat{\theta}$ is the MLE of the variance σ^2 then $\sqrt{\hat{\theta}}$ is the MLE of the standard deviation σ .

Maximum Likelihood Estimators II

- **Likelihood principle:** Consider two sets of data sampled from the same population. If the ratio of the likelihoods for each data set, $L_1(\theta, x)/L_2(\theta, y)$, does not depend on θ , then both data sets contain the same information about θ and should provide the same estimate of the parameter θ .

This principle emphasizes that the likelihood function is what informs us about the parameter θ based on observed data, and thus different data sets (with the same underlying population) should lead to the same conclusion regarding the value of θ .

Least Squares Estimators

- Now, a slightly different setup: Data are independent but possibly not from the same distribution.
- We observe Y and U and assume that the relationship between them is given by

$$Y = Q(u) + \varepsilon,$$

where Y is the dependent variable, u are the observations of an independent variable U , $Q(u)$ is some function of u describing the relationship between Y and U and ε is a random “error” with properties $E(\varepsilon) = 0$, $Var \varepsilon = \sigma^2$.

- Sometimes we may have several observations of Y for a given value of u , in some cases we have exactly one observation.
- The function Q is generally called the **regression** of Y on u .
- In general $Q(u) = \varphi(u, \theta_1, \dots, \theta_r)$, where φ is some function of u and unknown parameters $\theta_1, \dots, \theta_r$.
- When $Q(u)$ is linear in θ , for example $Q(u) = \alpha + \beta u$, we speak of a **linear regression model**.

Least Squares Estimators

- The method of least squares is based on finding values $\hat{\theta}_1, \dots, \hat{\theta}_n$, called **least squares estimate** (LSE), minimizing

$$S(\theta_1, \dots, \theta_r) = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \varphi(u_i, \theta_1, \dots, \theta_r))^2,$$

where y_{ij} is the j th observation of Y for the value u_i of U .

- The usual way to find LS-estimators of $\theta_1, \dots, \theta_r$ is by solving the set of so called **normal equations**

$$\frac{\partial S}{\partial \theta_i} = 0, \quad i = 1, \dots, r.$$

Recommended reading (Part 9)

For the lectures, please, read

- sections 11.7, 12.1, 12.2, 12.3 from Bartoszynski and Niewiadomska-Bugaj
- sections 9.1, 9.2, 9.3, 9.4, 10.1, 10.2, 10.3, 10.7 from Mittelhammer